1. Change Log

1) Should we add an additional conclusion to lemma 2.3 as in $s_b(k_{n+h}) = s_b(k_n) + s_b(k_h)$
2) Page 12, last line is not correct. With the notation of the premise of the proceeding lemma s is the largest integer s.t D is contained in an AP with step s. Therefore take $\theta = \frac{r}{s_0}$ then we have that, because we have that $D \subset \{n \equiv a \pmod{s}\}$ for some a therefore each d can be written as $d = ks + a$ so that $e(d\frac{r}{s_0}) = e(ks\frac{r}{s_0} + a\frac{r}{s_0}) = e(a\frac{r}{s_0})$ (constant).
3) Page 13, Shouldn't it be $b^j\alpha \longrightarrow \alpha_0 = \frac{a}{b-1} \mod 1$? Instead of $b^j\alpha \longrightarrow \frac{r}{b-1}$?
4) Page 14, "If $s > 1$, and $s \nmid d_0$ for some $d_0 \in D$, we turn to the Cantor set $C' = C'_{b/s,D'}$" it should be $C' = C'_{b,D'}$
5) Page 14, should we add a sentence to say that the uniform limit is obtained by switching from powers of δ to general M and N respectively in the same way it was done in (4.11)?
6) Page 15, "Thus, with $b^{n_0-1} < N < bn_0 \longrightarrow b^{n_0-1} < N < b^{n_0}$"
7) Proposition 5.1 has an error. Take $D \subset \{n \equiv r \pmod{q}\}$ for some $(r, q) = 1$ that would make $|\widehat{L}(\frac{a}{q})| = 1$ because

$$e((kq + r)\frac{a}{q}) = e(\frac{ra}{q}), \quad \textbf{(constant)}$$

8) Page 17, 3rd line from the bottom: "If $\gcd(a, a', b - 1) = g > 1$, it follows that for all integers k, b^k is congruent to 1, mod a and a'."
   This is incorrect. For example take $a = a' = 14$ and $b = 5$ then $\gcd(14, 14, 4) = 2$ but

$$b \equiv 5 \pmod{14}$$

9) Page 18, " If $\alpha = 0$ in (??), while $\beta \neq 0$, then the limit is zero, due to Proposition ??. " This builds on Prop 5.2 whose premise is incorrect therefore this has to change as well.
10) Page 19, Theorem 6.2,

$$\lim_N \mathop{\mathbb{E}}_{k\in C\cap[0,N]} (U, V)^{p(k_n,\sigma(k_n))}x \quad \text{exists in norm}$$

should be

$$\lim_N \mathop{\mathbb{E}}_{k\in C\cap[0,N]} (U, V)^{p(k,\sigma(k))}x \quad \text{exists in norm}$$

11) Page 19, By the spectral theorem we have that

$$U^k = \int_{a_1\in\mathbb{T}} e(ka_1)\mathrm{d}A_U(a_1), \quad V^{s_b(k)} = \int_{a_2\in\mathbb{T}} e(s_b(k)a_2)\mathrm{d}A_V(a_2)$$

$$\Rightarrow \mathop{\mathbb{E}}_{k\in C\cap[0,N]} U^kV^{s_b(k)}x = \int_{(a_1,a_2)\in\mathbb{T}^2} \mathop{\mathbb{E}}_{k\in C\cap[0,N]} e(ka_1 + s_b(k)a_2)\mathrm{d}A_U(a_1)\mathrm{d}A_V(a_2)$$

I dont understand what is

$$\lim_N \mathop{\mathbb{E}}_{C\cap[0,N]^2} e((m + n)/q)$$

Under the conditions of Theorem 6.1

$$\lim_N \mathop{\mathbb{E}}_{k\in C\cap[0,N]} e(ka_1 + s_b(k)a_2)$$

exists for all a_1, a_2 and is non-zero only when $sa_1 = \frac{h}{b-1} + \frac{r}{b^t}$ and $sa_2 = -\frac{h}{b-1}$ for integers h, r.

12) Page 11, The limit under discussion is

$$\lim_{N\to\infty} \mathop{\mathbb{E}}_{[N]\cap C} e(k\alpha + s_b(k)\beta)$$

however on the bottom of page 13 the limit

$$\mathop{\mathbb{E}}_{[N]} e(c_n\alpha + s_b(c_n)\beta)$$

is evaluated. The limit one wants, since the interest below resides at discussing the uniform distribution of the sequences c_n etc is , due to Weyl's criterion,

$$\lim_{N\to\infty} \mathop{\mathbb{E}}_{[N]} e(c_n\alpha + s_b(c_n)\beta)$$

13) Lemma 5.2 second bullet point, page 17. It must be "then the asymptotic distribution of $C \mod q$ is that of $C \cap [b^j] \mod q$

Remark 1.1. In this remark we show that the $\gcd(q, s) = 1$ is equivalent to $\gcd(q, D - D) = 1$ where s is the largest integer such that D lies in a progressions of step size s.

Under the aforementioned conditions, we can write for sequence $\{k_i\}_{i=1}^{|D|}$ and $0 \le r < |D|$

$$D = \{k_i s + r, i \in \{1, \dots, |D|\}\}$$

We will prove that

$$\gcd(k_{i+1} - k_i; i \in \{1, \dots, |D| - 1\}) = 1$$

Indeed if $g \coloneqq \gcd(k_{i+1} - k_i; i \in \{1, \dots, |D| - 1\}) > 1$, then since $d_{i+1} = d_i + (k_{i+1} - k_i)s$ we have that $D = \left\{d_1, d_1 + \widetilde{(k_2 - k_1)}(gs), \dots, d_1 + \widetilde{(k_{|D|} - k_{|D|-1})}(gs)\right\}$ which implies that D lies in a progression of step size $> s$ which is absurd. Therefore $\gcd(q, D - D) = \gcd(q, s)$ which clearly implies the desired conclusion.

1.2. **Other changes (Alex).**

(1) Changed SAAVEDRA-ARAYA reference to the published form
(2) Original Theorem 5.3 was incorrect. There are a lot of faulty conclusions. We removed.
(3) Removed $\pi_q(0) > 0$ from Lemma 5.2 and proved in standalone new Lemma 5.3.

1.3. **Revision of October 2026 (MTL).**

(1) **Standing assumption.** Throughout, $D \subset \{0, \dots, b-1\}$, $0 \in D$, $2 \le |D| < b$, and $C = \{0 = k_0 < k_1 < \cdots\}$. Signed digit sets are used only in the new §3.1 (power savings). $s = \gcd(D)$ is defined once in §2.3.
(2) **Prop. 2.2.** Index ranges corrected ($i \ge 1$, $n \ge 0$, $0 \le j < |D|^i$); part (iii) removed (unused).
(3) **Removed** the van der Corput inequality, Lemma `l:Diff`, and the Kronecker/weak mixing paragraph (no longer used).
(4) **§3.** Intersective and vdC definitions exclude 0. Equivalent formulations of vdC corrected; the false claim that intersective sets contain IP sets removed. New Question on power savings under the standing assumption. Proof of Prop. `Powersavings`

rewritten (off-by-one, notation); remark on difference sets corrected; new remark: power savings ≤ dimension.

(5) **Thm. `t:polysIntersective`.** Stated for $d^*(A) > 0$; $\mathcal{H}_{\text{rat}}$ uses $k \geq 1$; modulus $t' = tL$ to handle integer valued polynomials; lim inf and final step added; comparison with Bergelson–Furstenberg–McCutcheon (IP*) added.

(6) **Lemma `l;irrational`.** Restated with "exceptional" pairs; old case (3) removed; proof rewritten, including the passage to general windows $M \leq n < N$ and the uniform limit for all non-exceptional pairs.

(7) **Thm. `t:polyUniform`.** PUDU definition corrected (scalar polynomials). The van der Corput step replaced by a Cauchy–Schwarz argument over digit blocks, with new Lemma `l:badDirections`; this handles e.g. $p = \alpha y^2$, where the old step failed.

(8) **§5.** Prop. `p;sumDigits` and Lemma `l:Allq` restated and reproved (uniform on $g\mathbb{Z}_q$; product structure via CRT). Lemma `lem:positiveDensity` now follows from Lemma `l:Allq`. Joint modular theorem labelled `t:modularJoint`, with sufficient and necessary conditions.

(9) **§6.** Thm. `t:weaklyMixingPET` renamed `t:polyMeanErgodic`, proof completed. Height discussion, toy maximal problem and figure removed; pointwise convergence now cites Brokering Pinilla–Iosevich–Krause.

(10) **§7.** Digits $\leq (b-1)/2$; energy identity $E(C_{b^s}) = E(D)^s$; $\epsilon(D)$ defined explicitly; statement of Aistleitner–Larcher–Lewko Thm. 1 quoted.

(11) **Abstract and introduction** rewritten to match the results; history section given a heading.

(12) **References.** Added Maynard (Invent. Math. 2019), Ruzsa, Bourgain, Aistleitner–Lachmann–Pausinger, Grepstad–Larcher, Bergelson–McCutcheon (Memoirs), Brokering Pinilla–Iosevich–Krause; fixed Mauduit 2006 and capitalization.

(13) **Files.** Removed unused jointErgodic, sumOfDigits, Weyl, UniformDistribution_OldFile.